An integrated VigilODE polyalgorithm: Physical error composition, quotient stage compilation, and independent cumulative validation

VigilODE research synthesis
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We consolidate the numerical implementations and mathematical constructions of PR90–93 into a target-preserving solver design. The principal new connections are a selection-independent physical error induction with two distinct valid anchors, an output-specific quotient compiler for complete affine raw-U stage systems, and a precision-aware near-inverse correction rule. The compiler reduces a requested endpoint to one polynomial of degree below the declared operator-relation degree, while an explicit original-target residual extends the construction to near structure, stored coefficients, and bounded nonlinear remainders. A finite-menu error allocator and complete-session continuation contract connect these local methods to the adapter. Two source-registered exact-rational programs pass nine compiler fixtures and six physical-error check groups, together with correction and refusal controls. These checks support the written derivations; they do not validate new binary64 kernels or measure native performance. The report also specifies how a separate PR can include and validate all four preceding PRs without waiting for their merger. Its coverage records 73 changed crate/workflow paths, preserves seven unresolved qualification tasks and prior negative controls, and distinguishes source inclusion, scientific verification, and release readiness.

I. SCIENTIFIC QUESTION AND CUMULATIVE SOURCE

The goal is to exploit VigilODE's polyalgorithm: retain several faithful realizations of the same numerical target, remove work when declared structure permits it, and select an admitted candidate under a common accuracy and cost contract. The intended character is derivative-free first, with explicitly supplied JVP/AD capabilities used selectively where justified. An action can be matrix-free and still use a derivative; neither a polynomial basis nor a callback name establishes derivative freedom.

The four source PRs retain their own reports and receipts[1–4]. The source inspected here is the cumulative PR93 commit
`384dba091a3d9e1aac0ecddec264958298239253`.

Exact ancestry checks establish the chain from the pre-conversation implementation through PR90, PR91, PR92 and PR93. This research performs the requested assumption alignment, constructive derivation, and computational connection, corresponding to R3/R5/R9 of PhysMath Research Loop 1.3.0. The requested future independent-PR handoff is an execution design, not a completed native solver campaign.

TABLE I. Prior contributions retained in the synthesis. Existing implementation and historical measurements retain their original source and qualification scope.

PR	Contribution already present	Integration role
90	Conservative dominated-comparator stopping; aligned LU scratch; into-buffer and borrowed-row operations; conditional error/work theory	Same-target selection and inexpensive execution primitives
91	Causal input/prefix reuse; two-channel structural stages; RHS-only cubic planning; checked-integer entry comparison	Reusable model algebra, original residual and physical-budget leaves
92	Charged structural failure and residual enclosure; scalar/rational roots; production configuration; physical-work repair; profiles and qualification plan	Arithmetic ownership, failure accounting, deployment and adverse controls
93	Fused Clenshaw; 1/12 cubic defect and rational decay; variable-step transport; continuation and residence constructions	Local proof ingredients and seventeen pending implementation alternatives

Earlier polynomial, conditioned-resolvent, and coupled-stage work is also reused. Joint Chebyshev/Laguerre actions, signed output-adjoint Laguerre certification, and guarded Leja/Chebyshev/Laguerre routing already exist. The coupled solver already shares one projected diagonal factorization across eight stages. The finite nilpotent causal majorant and the two-channel sharp adjoint norm are established project results. None is presented as a new algorithm

in this report. The current guarded polynomial provider is dense symmetric; an opaque matrix-free replacement needs its own action-error, domain, norm, and identity evidence.

In particular, PR93's fused Clenshaw identity changes five distinct degree-m actions to one fused-only action. Its Laguerre defect transfer uses the established $|L_k(x)| \le e^{x/2}$ bound[8]; Chebyshev uses a unit factor on a verified self-adjoint chart in $[-1, 1]$. This changes execution direction and its local defects, not merely the name of a certificate. Fixed polynomial preconditioning must retain its linearity contract[9].

The current production facade covers smooth forward identity-mass IVPs with endpoint observations. Its default remains persistent RHS-only DOPRI5. A declaration of stronger physical certification must be justified by the additional contracts below; a successful local-tolerance solve does not automatically acquire that guarantee.

II. ONE TARGET, SEVERAL ADMISSIBLE REALIZATIONS

Each request fixes its model and provider epoch, input state and time, observations, norm, domain, arithmetic, callback permissions, and mathematical target. Four targets must remain distinguishable:

1. an original shifted linear system, checked by its original residual;
2. a represented fused φ combination, checked by complete action error;
3. an original nonlinear stage/root, checked by residual, domain and branch;
4. a physical step, checked by temporal, algebraic, representation and clock errors under a valid propagation argument.

A polynomial-action admission is a useful leaf of a physical-step certificate, but it is not the latter. Similarly, a quotient-stage endpoint below has the same discrete Rosenbrock target while still needing a temporal-error argument. Private adapters around existing result types can enforce these connections; the synthesis does not require replacing the solver hierarchy with a new general certificate framework.

A. Physical induction independent of selector smoothness

Let the original ODE have a unique flow $\Phi(t, s, x)$ on a certified tube. Use one physical norm $\|\cdot\|_*$. The accepted times $t_0 < \cdots < t_N$ may depend on the computed history. Rejected attempts do not advance the physical state. At the realized accepted step, assume

$$\|\Phi(t_{n+1}, t_n, x) - \Phi(t_{n+1}, t_n, z)\|_* \le \Lambda_n \|x - z\|_* , \tag{1}$$

$$\|\widehat{y}_{n+1} - \Phi(t_{n+1}, t_n, \widehat{y}_n)\|_* \le d_n, \tag{2}$$

on the required tube, with $\Lambda_n, d_n \ge 0$.

Proposition 1. If E_0 bounds the initial physical error, an outward recurrence

$$E_{n+1} \ge \Lambda_n E_n + d_n \tag{3}$$

bounds every committed physical error. The candidate selector may be discontinuous or history-dependent.

Proof. Add and subtract the same exact flow at $\widehat{y}_n$ in $\widehat{y}_{n+1} - y_{n+1}$. Apply Eqs. (1) and (2), then induct along the realized accepted grid. The proof never evaluates the selector at a second perturbed input, so no selector derivative or Lipschitz constant is required. □

Unrolling gives a product-weighted sum of local defects. For an unknown future adaptive grid those future products are an a posteriori expression, not an online budget reservation. Prefix accuracy also does not guarantee termination, adequate remaining resources, or discovery of another admitted candidate. Failed probes need transactional rollback or an explicit state-change defect; their work remains charged even when their physical state is discarded.

One concrete RHS-only route to d_n uses a differentiable reconstruction $v(t)$ and residual $R(t) = v'(t) - f(t, v(t))$. In a fixed Hilbert norm, suppose

$$\langle f(t, x) - f(t, z), x - z \rangle \le \mu(t) \|x - z\|^2 . \tag{4}$$

The energy inequality gives $\Lambda_n = \exp(\int_{t_n}^{t_{n+1}} \mu)$ and, for matching reconstruction endpoints,

$$d_n \le \int_{t_n}^{t_{n+1}} \exp\left(\int_s^{t_{n+1}} \mu(r)\, dr \right) \|R(s)\|\, ds. \tag{5}$$

The derivative of a polynomial reconstruction is algebraic; it requires no Jacobian callback. Rigorous integration still requires a valid residual and domain enclosure, not a few residual samples. Ranocha–Giesselmann give the relevant one-sided Lipschitz estimate and distinguish reliability of an error estimator from stability of its step-size controller[5]. Adjoint methods provide another global-error/conditioning perspective[6], but their general derivative-dependent adjoint machinery is not assumed to be available from a derivative-free provider.

B. Two valid anchors and a concrete invalid combination

Let $\Psi_{n,b}$ be the exact-arithmetic discrete recipe selected for this step, with its stored coefficients, frozen carrier, chart and norm held fixed. Suppose

$$\|\widehat{y}_{n+1} - \Psi_{n,b}(\widehat{y}_n)\|_* \le a_n. \tag{6}$$

There are two valid sufficient contracts:

$$\begin{aligned} &\tau_n^{\mathrm{num}} \ge \|\Psi_{n,b}(\widehat{y}_n) - \Phi(t_{n+1}, t_n, \widehat{y}_n)\|_*, \\ \text{F:}\quad &E_{n+1} \ge \Lambda_n E_n + a_n + \tau_n^{\mathrm{num}}, \end{aligned} \tag{7}$$

$$\begin{aligned} &\tau_n^{\mathrm{exact}} \ge \|\Psi_{n,b}(y_n) - \Phi(t_{n+1}, t_n, y_n)\|_*, \\ \text{D:}\quad &E_{n+1} \ge L_n E_n + a_n + \tau_n^{\mathrm{exact}}. \end{aligned} \tag{8}$$

For D, L_n is a discrete-map Lipschitz bound between the exact and numerical inputs. For F, Λ_n is the exact-flow stability factor. The two proofs insert different intermediate points. In particular, the selected carrier must not silently be rebuilt at y_n while calling the result the same function $\Psi_{n,b}$.

Take $\Phi(x) = x/2$, $\Psi(x) = x$, $y_n = 0$, and $\widehat{y}_n = e > 0$. There is no implementation error and the exact-input truncation is zero. The next physical error is e. Combining the flow factor $1/2$ with that zero truncation would give the false bound $e/2$. D uses $L_n = 1$ and is exact; F includes $\tau_n^{\mathrm{num}} = e/2$ and is also exact.

This matters for PR93's cubic planner. Its $1/12$ local estimate and rational decay sum are evaluated along the *exact* decaying trajectory. Its valid propagation uses nonexpansiveness of the midpoint discrete map, D with $L_n = 1$. Adding a stronger exact-flow contraction while leaving that truncation term unchanged is not justified. The counterexample identifies a possible integration mistake; it does not refute the existing PR93 proof.

III. ERROR OWNERSHIP, PHYSICAL NORMS AND CLOCKS

A useful certificate is a path between explicit physical-space anchors:

$$x^{(0)} = \Phi(t_{n+1}, t_n, \widehat{y}_n), \quad x^{(1)}, \dots, x^{(r)} = \widehat{y}_{n+1}. \tag{9}$$

If every edge has a bound $\left\|x^{(j+1)} - x^{(j)}\right\|_* \le \varepsilon_j$, telescoping gives $d_n \le \sum_j \varepsilon_j$. Model transformation, clock conversion, temporal method, algebraic solve and endpoint assembly are possible edges. A coarse certificate covering several consecutive edges is an alternative cover of that subpath. It is not an additional mandatory charge on top of the same individual edges.

For example, PR92's represented structural residual already encloses computed RHS, mean and lift errors. A subsequent residual-to-endpoint map transports those errors; it need not add them again as though absent. Endpoint accumulation outside that certificate still requires its own allowance. Conversely, a single direct physical-flow defect may cover the entire path.

For a reconstruction $\mathcal{R}$ with Lipschitz bound $K_{\mathcal{R}}$, a coordinate discrepancy η contributes at most $K_{\mathcal{R}}\eta$ plus separate reconstruction arithmetic. In the weighted infinity norm $\|x\|_w = \max_i |x_i|/w_i$, $w_i > 0$,

$$\|x\|_\infty \le \max_i w_i \, \|x\|_w, \qquad \|x\|_v \le \max_i \frac{w_i}{v_i} \, \|x\|_w. \tag{10}$$

The constants are sharp by concentrating on a maximizing component. A changed norm epoch therefore needs the appropriate conversion or a new physical evaluation. Identical scalar types do not identify physical norms.

Two clock conventions are consistent. Under the physical-label convention, the canonical time is the exact real value of stored t_n, so any difference between engine h_n and $t_{n+1} - t_n$ belongs to the local defect. Under the internal-clock convention, exact internal times τ_n are distinct from reported labels t_n. If $\|y'(s)\|_* \le M_n$ between the two times,

$$E_n^{\mathrm{obs}} \le E_n^{\mathrm{int}} + M_n |t_n - \tau_n|. \tag{11}$$

This observation-only displacement is not reinjected into every later internal recurrence. For $y' = 1$ with exact internal states, the observation error is the current label gap, not the sum of earlier gaps. For a nonautonomous shifted vector field, endpoint speed alone does not enclose the whole shifted step; time regularity and the corresponding sensitivity or exact-flow estimate are also needed.

IV. COMPILE COMPLETE AFFINE STAGES IN A FINITE OPERATOR ALGEBRA

The next result connects the existing conditioned resolvent to the whole causal stage target. Let s be the stage count, $s = 8$ in the current consumer, and let $\tau = h\gamma$, $h, \gamma > 0$. With the actual represented tableau, consider the autonomous/frozen-time raw-U equations

$$WU_i = \tau f\left(y + \sum_{j<i} a_{ij}U_j\right) + \gamma \sum_{j<i} c_{ij}U_j, \qquad W = I - \tau J_0. \tag{12}$$

A different mass matrix, stage time or partial-time term requires its matching equation; Eq. (12) is not changed implicitly.

Declare one whole-space chart and relation in a fixed induced norm:

$$Z = \tau J_0 = cI + \sigma C, \quad \sigma > 0, \quad \|C\| \le M, \quad \|\pi(C)\| \le \rho, \quad \deg \pi = d \ge 1. \tag{13}$$

The provider identity includes this chart, relation, norm and model epoch. Sampling a few vector actions does not establish Eq. (13). Set $z(x) = c + \sigma x$, $w_0(x) = 1 - z(x)$ and $\zeta = (1-c)/\sigma$. If $\pi(\zeta) \neq 0$, the existing quotient inverse is

$$v(x) = \frac{\pi(\zeta) - \pi(x)}{\sigma\pi(\zeta)(\zeta - x)}, \qquad w_0(x)v(x) = 1 - \frac{\pi(x)}{\pi(\zeta)}. \tag{14}$$

It has degree $d-1$. For $\rho = 0$, $v(C) = W^{-1}$. More generally, $\theta_\pi = \rho/|\pi(\zeta)| < 1$ yields

$$\left\|W^{-1}\right\| \le \frac{\|v(C)\|}{1-\theta_\pi} \le \frac{\sum_{k<d} |v_k| M^k}{1-\theta_\pi}. \tag{15}$$

The actual stored-polynomial preparation defect must be included in the native version, as in the existing conditioned implementation. A sensitive denominator can make this route unfavorable even when the original problem is solvable.

Theorem 2 (quotient stage compiler). Assume $\rho = 0$ and the declared affine identity $f(y+u) = f_0 + J_0 u$ on the stage domain. Define $B_{ij}(x) = a_{ij}z(x) + \gamma c_{ij}$ and degree-$< d$ representatives

$$r_i = \operatorname{rem}_\pi\left[v\left(1 + \sum_{j<i} B_{ij}r_j\right)\right]. \tag{16}$$

Then the unique solution of Eq. (12) and its endpoint are

$$U_i = \tau r_i(C) f_0, \qquad y_+ = y + \tau p_\beta(C) f_0, \qquad p_\beta = \sum_i \beta_i r_i, \quad \deg p_\beta < d. \tag{17}$$

Proof. Equation (14) makes v the inverse of w_0 modulo π. Evaluate Eq. (16) at C; every multiple of $\pi(C)$ vanishes. Multiplication by τf_0 gives the original affine stage equations. Invertibility of W and strict causality give uniqueness by stage induction. Applying the endpoint weights gives Eq. (17). Normality and diagonalization are not needed. □

After scalar preparation, an endpoint-only request uses at most $d-1$ full-space C actions by Horner evaluation, plus vector assembly and a base RHS if unavailable. Straightforward scalar preparation is $O(sd^2 + s^2 d)$ and is reusable at the same h/tableau/chart epoch. This is an operation bound, not measured speed. A requested embedded error or fixed dense-output linear functional gets its own polynomial. Several outputs can share one power sequence, but that power basis may have worse rounding than separate conditioned/Clenshaw evaluations. Complete error and total cost decide.

An endpoint-only implementation must not claim that it produced stage vectors. Dense/event consumers, stage diagnostics, FSAL or effectful callbacks may require more information or actual calls. Their contracts constrain the shortcut. The existing two-channel direct formula can be cheaper than this general compiler and remains a comparator, as does the shared projected factorization.

V. NEAR STRUCTURE AND STORED COEFFICIENTS PRESERVE THE ORIGINAL TARGET

For any actually stored degree-$< d$ polynomials r_i, form

$$s_i = w_0 r_i - 1 - \sum_{j<i} B_{ij} r_j = \alpha_i \pi + e_i, \qquad \deg e_i < d. \tag{18}$$

Because $\deg s_i \le d$, the quotient α_i is a *scalar*:

$$\alpha_i = -\frac{\sigma}{\pi_d}\left(r_{i,d-1} + \sum_{j<i} a_{ij} r_{j,d-1}\right). \tag{19}$$

For exact compilation $e_i = 0$. Outward coefficient arithmetic bounds e_i for stored coefficients, using $E_i(M) = \sum_{k<d} |e_{ik}| M^k$. This avoids creating an unnecessarily high-degree polynomial merely to certify each logical stage residual.

Let $\left\|\widehat{f}_0 - f(y)\right\| \le \varepsilon_0$ and define virtual stages $V_i = \tau r_i(C)\widehat{f}_0$ and $q_i = \sum_{j<i} a_{ij} V_j$. Assume a truthful nonlinear remainder bound

$$\|f(y+q_i) - f(y) - J_0 q_i\| \le \mu_i. \tag{20}$$

Their original raw-U residual is exactly

$$\begin{aligned} R_i = \tau\big[&(\alpha_i \pi(C) + e_i(C))\widehat{f}_0 + (\widehat{f}_0 - f(y)) \\ &- (f(y+q_i) - f(y) - J_0 q_i)\big], \end{aligned} \tag{21}$$

and hence

$$\|R_i\| \le b_i := |\tau|\left[(|\alpha_i|\rho + E_i(M))\left\|\widehat{f}_0\right\| + \varepsilon_0 + \mu_i\right]. \tag{22}$$

Let $\left\|W^{-1}\right\| \le \omega$ and suppose f is L_i-Lipschitz on the segments joining virtual and exact target stage arguments. The existing strictly causal majorant is

$$K_{ij} = \omega\big(|\tau| L_i |a_{ij}| + |\gamma c_{ij}|\big)(j<i), \qquad H = \omega \sum_{\ell=0}^{s-1} K^\ell. \tag{23}$$

Since $K^s = 0$, no global condition $\|K\| < 1$ is required. If the actual endpoint evaluation differs from $y + \sum_i \beta_i V_i$ by at most $\varepsilon_{\text{eval}}$, subtraction of original stage equations gives

$$\|\widehat{y}_+ - \Psi_h(y)\| \le |\beta|^T H b + \varepsilon_{\text{eval}}. \tag{24}$$

Here $\Psi_h(y)$ is the exact discrete stage target at the *same actual numerical input*. This is Eq. (6); temporal and clock errors remain separate under F or D. The proof is componentwise error $\le \omega b + K$ error followed by the finite sum.

The evaluation allowance includes output-coefficient assembly, basis conversion, operator action defects, recurrence arithmetic, multiplication by τ and endpoint addition. PR93's backward polynomial defect identities can be used here. Terms already enclosed in e_i or ε_0 keep their original owner.

Omitted RHS calls require a model-domain witness. A virtual argument radius is

$$\|q_i\| \le |\tau| \sum_{j<i} |a_{ij}| \left(\sum_{k<d} |r_{jk}| M^k\right)\left\|\widehat{f}_0\right\|. \tag{25}$$

For $f(x) = -kx - gx^3$, $g \ge 0$, and declared carrier $J_0 = -k$, a radius R gives the remainder bound $g(3|y|^2 R + 3|y|R^2 + R^3)$. This can be too loose, in which case evaluate the original RHS/residual or take another route. The Lipschitz domain also has to cover exact target stage arguments; a bound only on the virtual point does not establish it.

A. Signed transport of correlated structure defects

For exact coefficients and an affine model, $R_i = \tau\alpha_i\pi(C)f_0$. The residual blocks share one vector and need not be treated as independent. If C is self-adjoint in the declared Hilbert norm, with spectrum in a certified interval I on which $w_0 \neq 0$, let $L(x)$ be lower triangular with diagonal $w_0(x)$ and lower entries $-B_{ij}(x)$. Then

$$\sum_i \beta_i(V_i - U_i) = \tau g_\beta(C)\pi(C)f_0, \qquad g_\beta(x) = \beta^T L(x)^{-1}\alpha, \tag{26}$$

so the endpoint error is at most $|\tau|\rho\,\|f_0\|\sup_{x\in I}|g_\beta(x)|$. This follows from the affine block error equation and the spectral theorem. It is an alternative to Eq. (24), with stronger assumptions. A denominator enclosure and signed scalar-envelope preparation must be charged. For the two-projector model it reduces to the existing two scalar adjoints.

The adverse case $C = \left(\begin{smallmatrix}\delta & 1\\ 0 & -\delta\end{smallmatrix}\right)$, $\pi(x) = x^2$, has $\pi(C) = \delta^2 I$. Treating that near relation as exact omits a nonzero original residual. Equation (21) exposes the missing term directly. Independently, a nonnormal matrix can have a narrow eigenvalue set without a useful spectral-theorem bound. Numerical-range polynomial bounds, when actually certified, can use the established $1+\sqrt{2}$ constant[7]; no such witness is inferred here.

VI. PRECISION-AWARE REFINEMENT WITH A DERIVED STOPPING BUDGET

Let $r_k = f - Ax_k$, $\|I - AP\| \le \theta < 1$ and $\|A\| \le a$. Represent the residual, correction and update by

$$\begin{aligned} \widehat{r}_k &= r_k + e_k, \quad d_k = P\widehat{r}_k + z_k, \quad x_{k+1} = x_k + d_k + u_k,\\ \|e_k\| &\le \eta_k, \quad \|z_k\| \le \zeta_k, \quad \|u_k\| \le \upsilon_k. \end{aligned} \tag{27}$$

Direct substitution gives

$$r_{k+1} = (I - AP)r_k - APe_k - A(z_k + u_k), \tag{28}$$

$$R_{k+1} \le \theta R_k + \kappa_k, \qquad \kappa_k = (1+\theta)\eta_k + a(\zeta_k + \upsilon_k). \tag{29}$$

No commutation between A and P is required. A sharper independent bound on $\|AP\|$ may replace $1+\theta$.

For uniform $\kappa_k \le \kappa$,

$$R_m \le \theta^m R_0 + \kappa\frac{1-\theta^m}{1-\theta}, \qquad F = \frac{\kappa}{1-\theta}. \tag{30}$$

When $0 < \theta < 1$ and $R_0 > \varepsilon > F$, the least sufficient correction count for this bound is the least integer m such that

$$\theta^m \le \frac{\varepsilon - F}{R_0 - F}. \tag{31}$$

Outward integer verification decides the boundary. If $0 < \theta < 1$, $R_0 > \varepsilon$ and $\varepsilon \le F$, repeating this uniform bound cannot certify a finite success. This limits the chosen enclosure, not every algorithm or representable solution. If $\theta = 0$, zero corrections suffice when $R_0 \le \varepsilon$; otherwise one correction is certified when $\kappa \le \varepsilon$, including equality. If $\kappa > \varepsilon$, repetition of that same uniform bound does not establish success.

The exact example $\theta = 1/4$, $a = 1$, $\eta = 1/10000$, $\zeta = \upsilon = 1/20000$ has $F = 3/10000$. With $R_0 = 1$ and $\varepsilon = 1/1000$, Eq. (31) selects six corrections. The registered check includes a scalar construction attaining the bound for twelve updates, a noncommuting matrix example, and the $\theta = 0$ equality case.

Standalone refinement may use varying certified P_k, with the corresponding product-weighted sum of θ_k and κ_k. This does not justify a changing preconditioner inside an outer algorithm assuming fixed linearity; freeze that plan or use a separately qualified flexible method. The current native defaults `maximum_relation_ratio` $= 0.5$ and `stagnation_ratio` $= 0.95$ are operational policies. The derived rule can determine mathematical feasibility and the required work without treating those numbers as theorem constants. Explicit resource caps, failure charging and original-residual-controlled commit remain.

VII. ALLOCATE THE REMAINING BUDGET WITHOUT A FIXED ARBITRARY SPLIT

For a fixed request and valid tube, let component i have finitely many admitted options j, errors $e_{ij} \ge 0$ and costs $c_{ij} \ge 0$ in one declared unit. Let $W \ge 0$ map component errors into required observation bounds, r contain

already-owned errors, and ϵ be the budgets. The planner solves

$$\begin{aligned} &\min_{j_1,\dots,j_s} \sum_i c_{ij_i}, \\ &\text{subject to} \quad r + W(e_{1j_1}, \dots, e_{sj_s})^T \le \epsilon. \end{aligned} \tag{32}$$

Extend partial error/cost labels one component at a time. A label can be removed when another has no greater cost and no greater error in every observation coordinate, with at least one strict inequality. Equal labels must retain a representative. Completing a dominated label and its retained replacement with the same future options preserves dominance, proving that the least-cost feasible final label solves Eq. (32).

Shared setup changes this rule. Labels must carry a continuation state that determines future admissible choices, errors and costs. Prune only within that state, or under a proved simulation relation. A cold label costing zero now and 100 later does not dominate a warm label costing one now and zero later. Equal numerical errors do not make different factorization caches, coefficient plans or sessions interchangeable.

For weights $(2, 1)$ and budget $3/10$, let the error/cost menus be

$$\begin{aligned} A &: (e, c) \in \{(1/10, 4), (1/20, 7)\}, \\ B &: (e, c) \in \{(1/5, 1), (1/10, 3), (1/20, 6)\}. \end{aligned}$$

The exact optimum has error $3/10$ and cost 7. An equal physical split requires cost 10. These are declared toy work units, not native VigilODE measurements.

A continuous initializer is available for a stated work model $C_i(\eta_i) = a_i \log(B_i/\eta_i)$, $a_i, B_i > 0$, with $\sum_i w_i \eta_i \le B$. At an interior optimum, strict convexity and stationarity give

$$\eta_i = \frac{B a_i}{w_i \sum_j a_j}. \tag{33}$$

For positive bounds $\ell_i \le \eta_i \le u_i$, first check $B \ge \sum_i w_i \ell_i$. Take all upper bounds if they fit; otherwise use the clipped multiplier solution in its valid logarithmic range. The actual integer degree/iteration still needs its original outward test and finite comparison. A measured cost table makes this optimum conditional on that table; hardware latency is not a numerical-analysis constant.

An upper arithmetic allowance consuming the entire budget establishes infeasibility of that decomposition, not a universal precision impossibility. A true representability obstruction needs a lower distance: on the fixed lattice $(1/8)\mathbb{Z}$, the nearest value to $1/3$ is $3/8$, at distance $1/24$. Tolerance $1/100$ is impossible there. A loose upper allowance of one for the exactly representable value $1/2$ proves nothing comparable.

VIII. SESSION CONTINUATION AND COMPLETE WORK CONSERVATION

Three reuse mechanisms require different invalidation rules. Storage capacity may survive many target changes after initialization. Algebraic objects need their stated operator, step, chart, norm, model and arithmetic identities. Proof reuse needs its valid domain and anchor; a theorem can permit semantic reuse across different input bytes. These mechanisms should not be controlled by one undifferentiated cache-hit boolean. In particular, keep root workspace capacity while invalidating Anderson history when its map changes.

The current switching probe extracts an endpoint and proposed step inside `with_session`, then returns from the closure. The native session ceases to exist. Its existing state already includes workspace, controller, FSAL, complete output schedule, cursor, cumulative attempt budget and terminal state. Retaining that object can avoid rebuilding the winner.

Proposition 3 (retained-session continuation). For a deterministic transition system, pausing an execution at an accepted checkpoint and then continuing the same full state under the same callback response law and external controls preserves subsequent states, decisions, outputs and native callback trace. Work is the sum of the two disjoint trace segments.

Proof. The first states after the pause are identical; induction over future transition inputs gives identical states and emitted events. □ This does not follow from copying (t, y, h), and it does not claim identical host bookkeeping or elapsed time. Provider state affected by callbacks must be included or isolated; diagnostic counters can remain charged.

The schedule is a concrete integration condition. A session may land only at registered observations. A winning-capable probe should therefore receive the complete intended remaining schedule at construction and continue inside

the same closure. Adding a new earlier landing can change the first-window trace and needs a new comparison. Reaching an endpoint and retaining enough global attempt budget to continue are separate states.

For requested invocations A, admitted entries E, pre-entry refusals R, completed entries C, failed entries F, and pending entries P,

$$A = E + R, \qquad E = C + F + P. \tag{34}$$

All scheduled workers must be joined/accounted before setting $P = 0$. These are not physical callback identities; each entered callback/action has its own ledger. Thus Q-FAIR-AB's attempted-versus-admitted distinction and PR92's entered-versus-charged physical-work repair are different obligations.

A completed admissible competitor of cost C dominates an unfinished trial whose proved final-cost lower bound is already $L \geq C$, when ties prefer the completed result. For integer costs strict improvement is equivalent to final cost at most $C - 1$. A fixed nonnegative linear work functional supplies a monotone accumulated lower bound. This is a same-window statement and does not predict unexecuted fatal callbacks or future-regime costs. RHS counts, instructions and elapsed seconds remain different units. PR93's residence theorem still requires a future variation certificate; session reuse does not supply that missing assumption.

IX. RECORDED SUPPORTING CALCULATIONS

Both checker programs and their protocol were published before execution at `af7f3e9f55d652554042d2386bf07abb7d857d75`.

The two exact-rational programs exited successfully. Source/input hashes, stdout, stderr and commands are preserved in their receipts. A preceding Wolfram KKT calculation remains exploratory; its symbolic-variable warning is retained. One bounded complementary review clarified the zero-contraction boundary, strict frontier dominance, bounded-log feasibility and the cubic sign assumption before source registration. No recursive review was performed.

TABLE II. Nine compiler checks. n, s and d denote dimension, stage count and annihilator degree. The fixtures use small synthetic rational tableaus; they do not test native RODAS coefficients or floating arithmetic. Errors and causal infinity-norm bounds are rounded here for presentation; admission uses the complete exact rationals in the JSON output. Every row passed.

Fixture	n	s	d	Endpoint error	Causal bound
Exact scalar	1	3	1	0	0
Exact nonnormal nilpotent	2	8	2	0	0
Exact three-channel relation	3	3	3	0	0
Near annihilator	2	3	2	$1.201448\,10^{-4}$	$1.915502\,10^{-4}$
Self-adjoint signed-defect check	2	3	2	$1.201448\,10^{-4}$	$1.335575\,10^{-4}$
Ignoring structure defect: adverse control	2	1	2	$1.230769\,10^{-4}$	$1.846154\,10^{-4}$
Stored coefficient defect	2	3	2	$3.333333\,10^{-6}$	$7.038578\,10^{-6}$
Near-affine original target	2	3	2	$9.908213\,10^{-9}$	$1.464143\,10^{-8}$
Base-RHS representation defect	2	3	2	$3.988874\,10^{-5}$	$5.949902\,10^{-5}$

The self-adjoint fixture separately checks the signed identity and its Hilbert-norm inequality; the displayed bound is its causal infinity-norm bound. Additional algebra checks cover denominator refusal, a noncommuting right defect $1/20$, the minimum correction count six, varying certified correction plans and the $\theta = 0$ floor equality.

No native solver tests, benchmarks, profiles or new binary64 kernels were executed in this theory supplement. PR93's prior $5m \to m$ fused action and $512 \to 256$ cubic planning example retain their original mathematical scope. Neither those counts nor the present $d-1$ compiler bound are measured wall-time speedups. Existing negative controls remain useful: nonnormal spectral traps, wrong-root residual ambiguity, the unstable product of individually stable Verlet steps, unsuccessful polynomial scaling, and the historical W-filtered cost failure.

X. AN INDEPENDENT PR THAT VALIDATES THE CUMULATIVE IMPLEMENTATION

The separate validation branch should start from the final delivered integration commit and target `main`, after recording main's then-current full SHA. The observed main in this research is `8d0c79184e09efb5bdadc24a6315c60a71a44264`.

TABLE III. Six physical-error check groups, all passing exact assertions. These finite examples support the general written proofs.

Group	Discriminating exact result
Selection induction	Eight selected steps satisfy the bound; a positive defect family saturates its product-weighted sum
Two anchors	Actual error and each valid bound 1/5; invalid mixed bound 1/10
Representation and clock	Coordinate norm1 versus physical norm1000; current clock gap 1/100 stays observation-only
Finite menu	Exhaustive and pruned optimum cost7 versus equal-split cost10; warm-state total1 versus cold-state total100
Continuous allocation	$(\eta_1, \eta_2) = (1/16, 3/8)$; both KKT multipliers equal8
Precision distinction	Lattice distance 1/24; loose upper enclosure does not imply impossibility

It is an ancestor of the inspected cumulative source. No earlier PR merger is required to include their content. Reapplying nested PR heads by cherry-pick would duplicate already inherited changes. If main advances or diverges, reconcile it in the new worktree and qualify the resolved tree, not an old head.

The conversation range has 73 unique changed crate/workflow paths. Per-PR counts are 13, 20, 44 and 0 for PR90–93; overlap explains their sum. The machine-readable coverage maps every path to semantic gates and source hashes. Three scopes must be reported separately: source inclusion by ancestry, current-tree behavior covering the conversation's changes, and inherited/release qualification. Ancestry alone does not prevent a later revert or prove behavior. The main-target diff also contains substantial older work, which is not newly attributed to this conversation.

A. Current remote failures and the positive path forward

PR93's fresh-clone run 40609485096 passed its default dependency graph, workspace compilation and touched default contract. Integrator Clippy then failed at the same 23 sites already diagnosed by PR92. Later vendor/offline stages were skipped. Q-LINT therefore includes completing those stages after the actual lint repair; default build success is not offline qualification.

The authority run 40609485121 passed checker unit tests and append-only research-node validation, then failed the full authority scan. Its 10,133 citations and 797 distinct object strings include 53 missing, 27 missing allowlisted, 220 non-commit and 497 reachable classifications. The workflow uses full-depth checkout, so the failures cannot all be called shallow history. Julia package tree hashes, repository-qualified external commits, and published equal-tree mappings have different identity semantics. Q-LEDGER needs typed resolution and explicit mapping evidence while preserving genuine unresolved authorities; a blanket allowlist is not the correction.

TABLE IV. Inherited qualification obligations remain open. The new plan supplies their cumulative scope and positive next actions, not passing native results.

Gate	Required action
Q-GIVENS	Export exact inputs and independently check every returned residual before changing the historical expected status; `Ok(3640)` alone is not evidence of false convergence
Q-FAIR-AB	Reconcile attempted/admitted/refused work and independent callbacks; keep this separate from the repaired physical-entry ledger
Q-BENCH	Qualify the actual SUNDIALS/compiler/MPI environment and comparator physics; version changes are recorded factors
Q-LINT	Repair actual sites, preserve numerical/API semantics, then complete the skipped vendor/offline stages
Q-API	Test exhaustive downstream literals and enum use against the relevant predecessors; choose compatibility or an explicit migration
Q-NEGATIVE	Preserve the historical $15/7 > 1.3$ failure and inaccurate Off arm; record expected-negative and other ignored tests with their actual outcomes
Q-LEDGER	Validate typed authority and equal-tree mappings without rewriting historical execution identities or hiding unresolved scientific source

The future PR needs whole-package gates and cross-PR interactions: winning session versus rejection and observations; cache reuse versus step/provider/norm epochs; refusal, callback failure and joined workers; physical budget

versus polynomial/root/lift/clock errors; distinct/same-vector and fused/column outputs; planner/backend factorial comparisons; exact/near structure and nonlinear remainders; and retained adverse cases. Test, measurement and actual release/LTO profiles must be distinguished for floating/branch-sensitive fixtures.

Baseline and candidate builds require separate target directories and source/ binary identity. Replays of old scripts that write beside themselves belong in a disposable mirror/worktree, with new outputs captured separately. The final tree needs fresh behavior checks even when historical receipts are preserved. Conversely, old expensive campaigns need not all be rerun when the unique behavior is already covered under an explicit identity/reuse relation.

The existing 23-node/34-edge DAG, eight native suites and correction budgets remain intact. Builds, native tests, timings and profiles share the existing execution lock. Old structural and adapter correction loops are already 3/3; new alternatives do not reset them. The positive deployment path remains release qualification, matched-accuracy hardware measurements and actual-client qualification. An independently reproducible branch also needs independent oracles: a full original-equation solve or exact/high-precision derivation, not merely the same routine run from a different branch.

XI. IMPLEMENTATION DECISIONS AND UNRESOLVED THEORY

The immediate implementation sequence is to qualify the cumulative existing code, then measure N01 fused action and N02 planner changes independently, followed by N04 retained-session continuation. The quotient compiler is a focused N03/N05 extension on declared low-degree models. Its remainder and domain providers precede general near-affine omission. N06 residence retains its future-variation obligation.

Allocation, scratch reuse, output fusion and complete-session lifetime can be improved through code while preserving their stated mathematical targets. Outward preparation and defect kernels require implementation-level arithmetic proofs and native checks. General nonlinear carrier order, stiffness-uniform accuracy, physical branch control, arbitrary domain certification and future cost variation require additional theory. No local failed candidate establishes a universal impossibility of the project's original construction.

The inherited references preserve these distinctions. Algebraic-geometry regularization and nonlinear root certification[10, 11] do not by themselves select the original physical branch or provide a derivative-free implementation. The DAE symmetry construction has its stated reduction and linear-system scope[12]. Geometric RK/composition identities[13] still require separate treatment of adaptive selection and inexact stages.

The integrated design therefore selects among eligible structures and targets, transports each admitted local result through a correctly anchored physical contract, and charges every attempted computation. Proof-derived constants determine admissibility and precision feasibility. Declared resource budgets and measured service costs determine the economic choice within that region. The complete derivations, exact outputs, provenance, coverage and handoff are stored under `research/integrated_polyalgorithm_20261028/`.

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